
ZETAPLEX DM

DEMINERALIZING AGENT / SEQUESTERING AGENT

Product Description:

Zetaplex DM is a highly effective, acidic demineralizing agent specifically formulated for pre-bleach demineralisation treatment of cellulosic and synthetic fibres. Designed as a preparatory step before the bleaching process, it delivers powerful complexing and sequestering action against heavy metal ions including calcium (Ca^{2+}), magnesium (Mg^{2+}), iron (Fe^{3+}), copper (Cu^{2+}), and other metallic impurities present in process water, chemicals, and natural fibres.

Heavy metal contamination prior to bleaching is one of the leading causes of catalytic damage to hydrogen peroxide, resulting in fibre degradation, patchy bleaching, loose spots, pinholes, and reduced whiteness. Zetaplex DM neutralises these risks by effectively chelating and removing such impurities before the bleach bath is introduced. Its acidic character makes it especially effective in the pre-bleach acidification step, while its excellent compatibility with acids and common process chemicals ensures seamless integration into existing pre-treatment sequences.

Key Benefits:

Dedicated Pre-Bleach Demineralisation: Specifically engineered as a demineralising step before bleaching — eliminates metallic impurities that would otherwise cause catalytic decomposition of hydrogen peroxide and damage fabric integrity.

Prevents Catalytic Fibre Damage: Removes iron and copper ions before the bleach bath, preventing catalytic attack on fibres — protecting against pinholes, holes, and fibre tendering caused by uncontrolled peroxide decomposition.

Superior Heavy Metal Chelation: Offers strong complexing action against Ca^{2+} , Mg^{2+} , Fe^{3+} , Cu^{2+} , and other heavy metal ions, keeping them in solution and preventing precipitation onto the fibre or machinery.

Enhanced Bleaching Performance: A well-demineralised fabric entering the bleach bath results in more uniform, brighter, and cleaner whites with consistent reproducibility across batches.

Acidic Pre-treatment Benefit: Its acidic nature aids in the neutralisation of residual alkali on the fabric from prior processing steps, creating an optimal surface condition before the bleaching sequence begins.

Scale and Deposit Prevention: Prevents build-up of calcium carbonate, iron hydroxide, and other mineral deposits on machinery surfaces, reducing downtime and extending equipment life.

Threshold Sequestering Effect: Maintains heavy metal ions in dissolved form even in supersaturated process liquors of CaCO_3 , CaSO_4 , BaSO_4 , and $\text{Fe}(\text{OH})_3$ at low dosage levels, preventing precipitation even under challenging water quality conditions.

Versatile and Compatible: Fully compatible with acids and common pre-treatment auxiliaries, enabling smooth incorporation into standard pre-bleach preparation recipes without process disruption.

Properties:

Chemical Constitution	Organically modified complexing agent based on organic acid chemistry
Ionic Character	Anionic
pH of 1% Solution	Acidic (below 7)
Appearance	Colourless to light straw yellow clear liquid
Physical Form	Liquid
Solubility	Freely soluble in water
Storage Stability	Stable under normal climate conditions; store away from freezing temperatures and direct heat
Stability to Acids	Excellent
Stability to Alkalis	Good
Compatibility	Compatible with acids and standard pre-treatment process chemicals

Applications & Dosage:

Prior to bleaching, fabric carries metallic impurities from several sources — feed water hardness, processing chemicals (salt, soda ash, caustic soda), or from the natural fibre itself. If not removed before the bleach bath, these metal ions cause:

- Catalytic decomposition of hydrogen peroxide leading to fibre damage, pinholes, and holes
- Uneven and patchy bleaching results
- Lower achieved whiteness despite high peroxide dosage
- White spots and surface defects on dyed or printed substrates
- Loss of reproducibility and increased rejection rates
- Scale deposits on bleaching machinery and heat exchangers

Demineralisation Bath — Exhaust Process (Pre-Bleach)

Function	Product	Dosage	Parameters & Conditions
Demineralizing Agent	ZETAPLEX DM	0.3–1.0 g/l	Auxiliary dosing – raise temperature to 70°C–85°C – hold for 20–30 minutes – drain bath – proceed to bleaching sequence
Wetting Agent	Zetawet SWLF CONC	0.3–0.5 g/l	

Note: Zetaplex DM is specifically recommended as the final preparatory step immediately before the bleaching bath. The acidic treatment at 70°C–85°C ensures thorough removal and complexing of heavy metals from both the fibre and the process water, creating ideal conditions for controlled, uniform peroxide bleaching with maximum whiteness and fibre safety.

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datasheet serves as a guideline, believed to be accurate. Users are advised to assess recommendations before actual application. Safety Data Sheet (SDS) is available upon request.